given secure access from anywhere on any device at any time.

- Because devices and data are stored in secured locations, even if employees lose their laptop or phone, there is nothing to steal from it, and so there is no security breach.
- Cloud workspaces make remote working much simpler. They help IT teams to support work from home.
- Collaboration and file sharing is much simpler as all the infrastructure is on the cloud.
- IT teams can spin up new cloud desktops faster. Users quickly get access to desktops and apps when they need them, making them more efficient and IT more agile.
- Availability is much better and disaster recovery faster with cloud workspaces. The underlying cloud platform offers availability sets, zones, and regional pairs that improve availability.
- Flexibility and scalability are two major benefits of cloud desktops, as they allow businesses, whether large or small, to grow at their own pace, adding and taking away licences as they're required.
- It's also much more straightforward to roll out updates for software or add new applications, without having to worry about updates and security fixes on individual machines.
- Setting up and scaling DevOps environments for the enterprise product teams is simpler and faster with cloud workspaces. Usage of readily available cloud services for development, testing, and analytics makes IT agile.
- Cloud vendors offer IaaS, PaaS, and SaaS services that enterprises can choose from based on their need, comfort, strategy, and policies.
- By unifying all of a business groups's cloud tools, IT departments can take advantage of a 'single pane of glass' approach to management and administration, ensuring that performance remains smooth. Integration between apps is seamless.
- With cloud workspaces, security, data governance and compliance can be easily enforced and monitored.

The challenges

Cloud desktops can be a challenge if the Internet connection is not reliable and patchy. In case the applications are resource-intensive, cloud workspaces can be costlier than on-premises in the long run. Other challenges include cloud vendor lock-in unless enterprises plan carefully and take a vendor-neutral approach. Enterprises should evaluate cloud costs on a continuous basis to optimise consumption, and should be aware of hidden charges. IT departments need to ensure that they avoid cloud sprawl, and keep their digital tools as unified and streamlined as possible.

Cloud workspace deployment

A cloud workspace is virtual desktop infrastructure that can be used in multiple ways:

- Greenfield public cloud deployment with Microsoft Windows virtual desktop, Windows 365, Amazon WorkSpaces, Google Workspace, etc.
- Migration of on-premises remote desktop services (RDS)/VDI deployments to cloud workspaces.
- Hybrid deployments by separating control plane and workload layers, making use of cloud service brokers such as Citrix Cloud, VMware Cloud, etc. Workload can be hosted in the cloud or even on-premises.

Table 1

Phase	Activity
Assessment	Identification of the requirements is an important step, based on which we can choose the right VM instance type, enabling optimisation with multi-session operating systems, persistence, access models, etc. The existing environment must be assessed to understand the user landscape, including personas, access methods and experience levels, data landscape, as well as the application and infrastructure landscape. Cloud vendors provide monitoring tools, the data trends from which can be used to find resource demands, application utilisation, etc.
Design	Design a secure and resilient cloud workspace environment, including networking, storage, virtual machine (VM) instances and security, based on business needs and requirements. Most of the cloud vendors provide well-architected frameworks — best practices that help in planning and designing the cloud workspaces.
Build	Build and deploy user persona based cloud workspace host pools and Windows 11/10/RDS images with base applications and application layering as per the approved design. Persona analysis is very important to size properly and to meet end user demands.
Migration	Plan and perform a phased migration to the new environment with pilot testing, change management and end user training.
Support	After successful deployment and migration, provide managed support to monitor and maintain the health of the solution as well as perform ongoing optimisation. Cloud vendors offer native services for monitoring, self-healing, self-service, alert triggers, security, and compliance. Enterprises can quickly customise these to suit their needs.